Huawei Access Node Routing Protocol Flap Resolution Procedure for Jayanagar Metro Site

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2778

Description

The failure domain of Routing Protocol Flap, specifically manifested through BGP route churn, OSPF adjacency loss, or ISIS LSP flooding, on the Huawei access node JAY-ACC-68 at Jayanagar Metro Site, necessitates a comprehensive understanding of the underlying protocol mechanics and hardware architecture. The Huawei access node, operating in the role of an access device, is critical for facilitating connectivity between the metro site and the broader network. The BGP route churn could be attributed to instability in the network, possibly due to misconfigured route reflectors or issues with the iBGP/eBGP sessions. OSPF adjacency loss might stem from layer 2 issues, such as Ethernet link failures or misconfigurations, affecting the formation and maintenance of OSPF neighbor relationships. ISIS LSP flooding could result from excessive LSP updates, potentially triggered by frequent changes in the network topology or configuration errors. The interplay between these routing protocols and the specific hardware and software configurations of the Huawei device dictates the approach to diagnosing and resolving these issues. Furthermore, understanding the nuances of Huawei's CLI dialect is essential for effective troubleshooting and configuration adjustments.

Pre-requisite Checks

- Verify the physical integrity of the access node and its connections at Jayanagar Metro Site.
- Check the status of redundant links to ensure no single point of failure.
- Confirm that all necessary personnel and equipment are available for potential on-site interventions.

Potential Root Causes

Cause	Probability
Misconfigured BGP route reflectors	High
Layer 2 issues affecting OSPF adjacency	Medium
Excessive ISIS LSP updates due to network topology changes	Low

Diagnostic Commands & Expected Outputs

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Command: display ip bgp peer, Expected Output: List of current BGP peers with their status, Description: This command is used to isolate issues with BGP peer connections, which could be a root cause of the routing protocol flap. By examining the output, one can identify if there are any BGP peers that are not in the established state, indicating a potential problem.

Command: display ospf peer, Expected Output: List of current OSPF neighbors with their status, Description: This command helps in
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diagnosing OSPF adjacency issues by displaying the current state of OSPF neighbors. It can reveal if there are any neighbors that are not in the full state, suggesting a possible issue with OSPF adjacency.

Command: display isis peer, Expected Output: List of current ISIS neighbors with their status, Description: Similar to the OSPF peer command, this command is utilized to troubleshoot ISIS neighbor relationships, which could be impacted by LSP flooding. The output will show the current state of ISIS neighbors, aiding in the identification of any issues.
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Resolution Steps

Step 1: Step: Adjust BGP Route Reflector Configuration, Command: bgp peer route-reflector-client, Description: If the diagnosis indicates issues with BGP route reflectors, this step involves adjusting the configuration to correctly reflect the route reflector setup. This ensures that routes are properly advertised and received within the BGP network.

Step 2: Step: Resolve Layer 2 Issues for OSPF Adjacency, Command: interface ip ospf enable, Description: For issues related to OSPF adjacency loss, ensuring that the OSPF protocol is enabled on the relevant interfaces is crucial. This step involves verifying and adjusting the configuration to enable OSPF on the necessary interfaces, thereby facilitating the formation of OSPF adjacencies.

Step 3: Step: Optimize ISIS LSP Updates, Command: isis timer lsp-generation , Description: To mitigate the impact of excessive ISIS LSP updates, adjusting the timer for LSP generation can help reduce the frequency of updates. This step involves configuring an appropriate interval for LSP updates to minimize the burden on the network while still ensuring topology changes are properly propagated.

Post-Check Verification

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Command: display ip routing-table, Expected Output: Current IP routing table, Description: After implementing the resolution steps, verifying the IP routing table is essential to ensure that routes are being correctly advertised and learned. This command displays the current state of the routing table, allowing for the confirmation of successful route propagation.

Command: display ospf routing-table, Expected Output: Current OSPF routing table, Description: Specific to OSPF, this command is used to verify that OSPF routes are being correctly calculated and inserted into the routing table. It helps confirm that OSPF adjacency issues have been resolved and that OSPF routes are being properly propagated.

Command: display isis routing-table, Expected Output: Current ISIS routing table, Description: For ISIS, verifying the ISIS routing table is crucial to ensure that ISIS routes are correctly calculated and inserted. This command provides insight into the ISIS routing table, confirming that ISIS LSP updates are properly handled and that the network topology is correctly reflected in the routing table.
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Roll-back Procedures

- Step: Revert BGP Route Reflector Configuration, Command: undo bgp peer route-reflector-client, Description: If the adjustment to the BGP route reflector configuration does not resolve the issue or introduces new problems, reverting this change is necessary. This command removes the route reflector client configuration for the specified peer, returning the setup to its previous state.

- Step: Revert OSPF Interface Configuration, Command: interface undo ip ospf enable, Description: In the event that enabling OSPF on an interface does not resolve the adjacency issue or causes unforeseen problems, it may be necessary to revert this configuration change. This command disables OSPF on the specified interface, restoring the previous configuration.

- Step: Revert ISIS LSP Generation Timer, Command: undo isis timer lsp-generation, Description: If adjusting the ISIS LSP generation timer does not mitigate the issue of excessive LSP updates or introduces new problems, reverting this change is required. This command resets the LSP generation timer to its default value, thereby restoring the original behavior of ISIS LSP updates.